Oct27-practice-read-data

This document is to practice reading data in R

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.4     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(tidyverse)

data_raw <- read.csv(
  "data/db_hiscod_csv_v1_en.csv",
  sep = ";",
  header = TRUE
)
```

Make a copy of the data

```
data_copy <- data_raw |>
  separate_wider_delim(
    cols = 1,
```

```
    delim = ";",
    names = strsplit(names(data_raw)[1], ";")[[1]]
  )
```

Take a look of the data

```
head(data_copy)
```

```
# A tibble: 6 x 85
  id    id_riot_hiscod id_riot_original_database title riot_date  year month_num
  <chr> <chr>          <chr>                     <chr> <chr>     <int>     <int>
1 1     HISCOD_00001   JN_0001_1789              Even~ 1789-03-~  1789         3
2 2     HISCOD_00002   JN_0002_1789              Even~ 1789-00-~  1789        NA
3 3     HISCOD_00003   JN_0003_1789              Even~ 1789-04-~  1789         4
4 4     HISCOD_00004   JN_0004_1789              Even~ 1789-03-~  1789         3
5 5     HISCOD_00005   JN_0005_1789              Even~ 1789-01-~  1789         1
6 6     HISCOD_00006   JN_0006_1789              Even~ 1789-03-~  1789         3
# i 78 more variables: month <chr>, day <int>, day_week <chr>,
#   riot_type_hiscod_num <int>, riot_type_hiscod <chr>,
#   riot_type_original_database_1 <chr>, riot_type_original_database_2 <chr>,
#   riot_type_original_database_3 <chr>, nb_participants <chr>,
#   women_participation <chr>, city_source <chr>, city_name <chr>,
#   city_code <chr>, code_insee_pre_2016 <chr>, city_latitude <dbl>,
#   city_longitude <dbl>, geo_precision <int>, country_name <chr>, ...
```

```
colnames(data_copy)
```

```
 [1] "id"                           "id_riot_hiscod"
 [3] "id_riot_original_database"    "title"
 [5] "riot_date"                    "year"
 [7] "month_num"                    "month"
 [9] "day"                          "day_week"
[11] "riot_type_hiscod_num"         "riot_type_hiscod"
[13] "riot_type_original_database_1" "riot_type_original_database_2"
[15] "riot_type_original_database_3" "nb_participants"
[17] "women_participation"          "city_source"
[19] "city_name"                    "city_code"
[21] "code_insee_pre_2016"          "city_latitude"
[23] "city_longitude"               "geo_precision"
```

```
[25] "country_name"                  "admin_level_1_type"
[27] "admin_level_1"                 "admin_level_2_type"
[29] "admin_level_2"                 "historical_political_entity"
[31] "historical_admin_level_1_type" "historical_admin_level_1"
[33] "historical_admin_level_2_type" "historical_admin_level_2"
[35] "description_event_english"     "description_event"
[37] "primary_sources_1"             "url_1"
[39] "primary_sources_2"             "url_2"
[41] "primary_sources_3"             "url_3"
[43] "primary_sources_4"             "url_4"
[45] "primary_sources_5"             "url_5"
[47] "primary_sources_6"             "url_6"
[49] "primary_sources_7"             "url_7"
[51] "primary_sources_8"             "url_8"
[53] "primary_sources_9"             "url_9"
[55] "primary_sources_10"            "url_10"
[57] "bibliography_1"                "pages_1"
[59] "bibliography_2"                "pages_2"
[61] "bibliography_3"                "pages_3"
[63] "bibliography_4"                "pages_4"
[65] "bibliography_5"                "pages_5"
[67] "bibliography_6"                "pages_6"
[69] "bibliography_7"                "pages_7"
[71] "bibliography_8"                "pages_8"
[73] "bibliography_9"                "pages_9"
[75] "bibliography_10"               "pages_10"
[77] "bibliography_11"               "pages_11"
[79] "bibliography_12"               "pages_12"
[81] "comments"                      "author"
[83] "contributor"                   "date_creation"
[85] "date_edition"
```

The column I am interested in are id, year, contry_name, women_participation. So I am slecting the three column to check the dataset.

```
data_select <-
data_copy |>
  select(id, country_name, year, women_participation)
```

For contry, I only want to include France; for year i want to range from 1700 to 1900

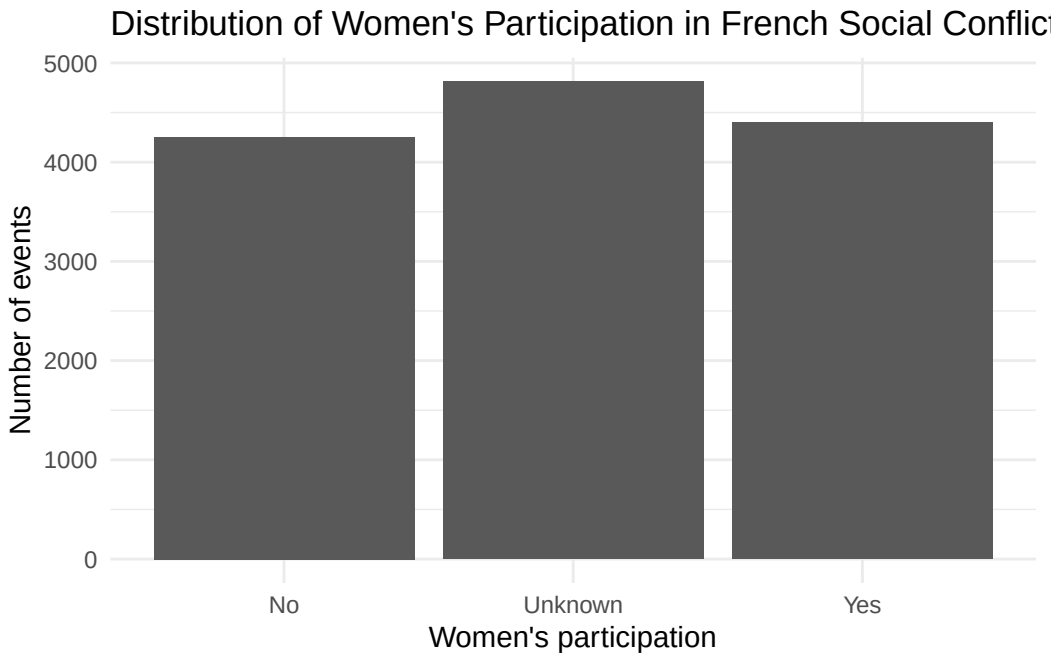
```
france_data <-data_select |>
    filter(
    country_name == "France",
    year >= 1700,
    year <= 1900
  )
head(france_data)
```

```
# A tibble: 6 x 4
  id    country_name  year women_participation
  <chr> <chr>        <int> <chr>
1 1     France        1789 Unknown
2 2     France        1789 Unknown
3 3     France        1789 No
4 4     France        1789 No
5 5     France        1789 Unknown
6 6     France        1789 Unknown
```

Check the distribution of women's participation

```
library(ggplot2)

ggplot(france_data, aes(x = women_participation)) +
  geom_bar() +
  labs(
    title = "Distribution of Women's Participation in French Social Conflicts",
    x = "Women's participation",
    y = "Number of events"
  ) +
  theme_minimal()
```



write a test for checking the country is France and the year range from 1700 to 1900

```
# Simulated dataset matching your actual data
test_data <- tibble(
  id = 1:4,
  country_name = c("France", "France", "Germany", "France"),
  year = c(1789, 1850, 1800, 1690),
  women_participation = c("Unknown", "No", "Yes", "Yes")
)

# Filter France and years 1700-1900
test_result <- test_data |>
  filter(country_name == "France", year >= 1700, year <= 1900)

# Test the filter
stopifnot(nrow(test_result) == 2)
stopifnot(all(test_result$country_name == "France"))
stopifnot(all(test_result$year >= 1700 & test_result$year <= 1900))
```